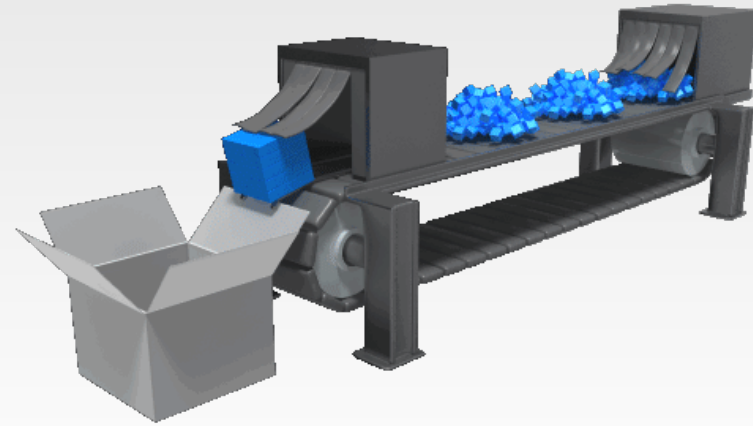




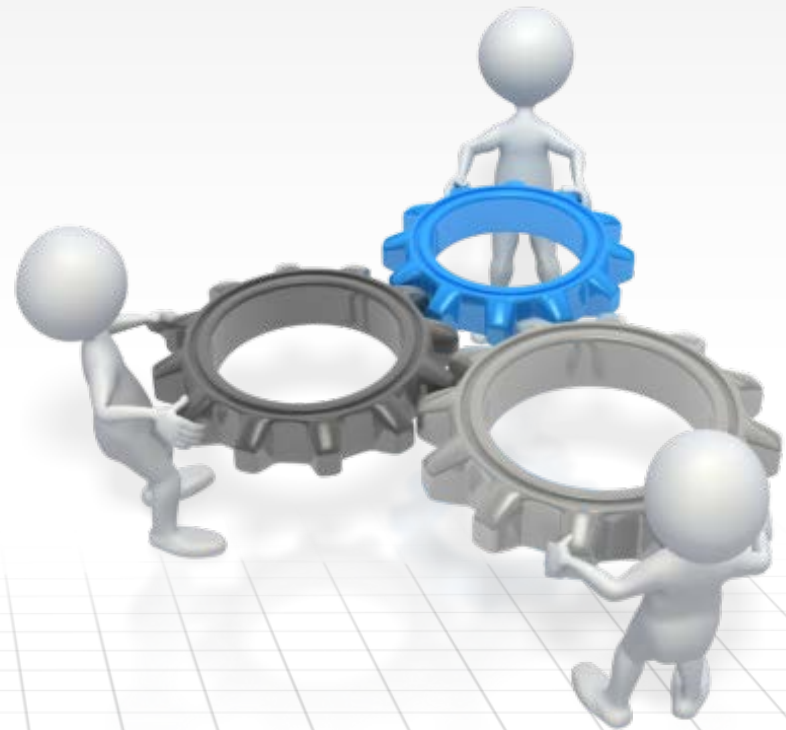
LECTURE 5

MATERIAL HANDLING

PART 2 - STORAGE SYSTEMS

Contents

1. Objectives
2. Review of material handling equipment
3. The principles of material handling
4. Storage systems
5. Conclusion



Objectives

1. The presentation of material storage equipment
2. Knowledge of the principles underlying the development of a material handling system
3. Analysis of material storage systems



Review

1. Material handling equipment
2. Material handling requirements





The Principles of Material Handling (1)

Principle 1. PLANNING PRINCIPLE: All material handling should be the result of a deliberate plan where the needs, performance objectives, and functional specification of the proposed methods are completely defined.

Principle 2. STANDARDIZATION PRINCIPLE: Material handling methods, equipment, controls, and software should be standardized within the limits of achieving overall performance objectives and without sacrificing needed flexibility, modularity, and throughput¹.



¹Throughput is defined as the hourly rate at which the storage system receives and puts loads into the storage and/or retrieves and delivers loads to the output station



The Principles of Material Handling (2)

Principle 3. WORK PRINCIPLE: Material handling work should be minimized without sacrificing productivity or the level of service required of the operation.

Principle 4. ERGONOMIC PRINCIPLE: Human capabilities and limitations must be recognized and respected in the design of material handling tasks and equipment to ensure safe and effective operations.

Principle 5. UNIT LOAD PRINCIPLE: Unit loads shall be appropriately sized and configured in a way which achieves the material flow and inventory objectives at each stage in the supply chain.

Principle 6. SPACE UTILIZATION PRINCIPLE: Effective and efficient use must be made of all available space.



The Principles of Material Handling (3)

Principle 7. SYSTEM PRINCIPLE: Material movement and storage activities should be fully integrated to form a coordinated, operational system that spans receiving, inspection, storage, production, assembly, packaging, unitizing, order selection, shipping, transportation, and the handling of returns.

Principle 8. AUTOMATION PRINCIPLE: Material handling operations should be mechanized and/or automated where feasible to improve operational efficiency, increase responsiveness, improve consistency and predictability, decrease operating costs, and eliminate repetitive or potentially unsafe manual labor.

Principle 9. ENVIRONMENTAL PRINCIPLE: Environmental impact and energy consumption should be considered as criteria when designing or selecting alternative equipment and material handling systems.

Principle 10. LIFE CYCLE COST PRINCIPLE: A throughout economic analysis should account for the entire life cycle of all material handling equipment and resulting systems.



Storage Systems



The function of a material storage system is to store materials for a period of time and to permit access to those materials when required.

Measures used to assess the performance of a storage system:

- ✓ storage capacity
- ✓ density
- ✓ accessibility
- ✓ throughput
- ✓ utilization
- ✓ reliability



The performance of a storage system in accomplishing its function must be sufficient to justify its investment and operating expense.



Storage Systems



Storage capacity is the available volume or number of storage compartments available.

Storage density is defined as the volumetric space available for actual storage relative to the total volumetric space in the storage facility.

Accessibility refers to the capability to access any desired item or load stored in the system.

Throughput is defined as the hourly rate at which the storage system receives and puts loads into the storage and/or retrieves and delivers loads to the output station.

Utilization is defined as the proportion of time that the system is actually being used for performing storage and retrieval operations compared with the time it is available.

Availability is a measure of system reliability, defined as the proportion of time that the system is capable of operating compared with the normally scheduled shift hours.



Storage Location Strategies



Strategies used to organize stock in a storage system:

- randomized storage
- dedicated storage

Dedicated storage is used when:

- items are stored in part number or product number sequence,
- items are stored according to activity level,
- items are stored according to their activity-to-space ratios.

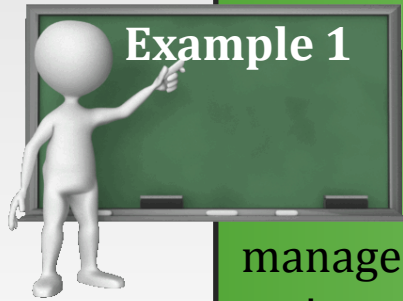


Less total space is required in a storage system that uses randomized storage

Higher throughput can be achieved when a dedicated storage strategy is implemented based on activity level.¹⁰



Storage Location Strategies



Example 1

Suppose that a total number of 50 types of parts must be stored in a storage system. For each type, average order quantity is 100, average depletion rate is 2 parts per day, and safety stock level is 10 units. Each part requires one storage location in the system. The management has scheduled incoming orders so that a different type of parts arrives each day. Determine the number of storage locations required in the system under two alternative strategies: randomized storage and dedicated storage.

Solution:

Maximum inventory level = $100 + 10 = 110$ parts

Minimum inventory level = 10 parts

Average inventory level = $(110 + 10) / 2 = 60$ parts

Number of storage location = $50 \times 60 = 3000$ locations (the case of randomized access)

Number of storage location = $50 \times 110 = 5500$ locations (the case of dedicated access)





Automated Storage Systems (1)

Automated storage systems divide into two general types:

- automated storage/retrieval systems (ASRS)
- carousel storage systems

The biggest difference between ASRS and carousel is in the construction of the equipment:

- The basic ASRS consist of a rack structure for storing loads and a storage/retrieval mechanism whose motions are linear.
- A basic carousel uses storage baskets suspended from an overhead conveyor that revolves around an oval/round track loop to deliver the baskets to a load/unload station.

Automated Storage Systems (2)

Feature	Basic ASRS (a)	Basic Carousel storage system (b)
Storage structure	Rack system to support pallets or shelf system to support tote bins	Baskets suspended from overhead conveyor trolleys
Motions	Linear motions of S/R machine	Revolution of overhead conveyor trolleys around oval or round track
Storage/retrieval operation	S/R machine travels to compartments in rack structure	Conveyor revolves to bring baskets to load/unload station
Replication of storage capacity	Multiple aisles, each consisting of rack structure and S/R machine	Multiple carousels, each consisting of oval or round track and suspended bins



(a)



(b)



Sizing the ASRS rack structure

- The total storage capacity of one storage aisle is:

$$CPA = 2 \times n_y \times n_z$$

where:

CPA – capacity per aisle

n_y – number of load compartments along the length of the aisle

n_z – number of load compartments that make up the height of the aisle

- Dimensions of a rack structure are calculated using next formulas:

$$W = 3 \times (x + a)$$

$$L = n_y \times (y + b)$$

$$H = n_z \times (z + c)$$

where: W – width of one aisle of the ASRS;

L – length;

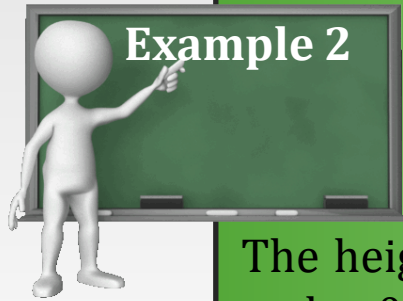
H – height

x,y,z – dimensions of the unit load (mm, in)

a,b,c – allowances designed into each storage compartment



Sizing the ASRS rack structure



Example 2

Each aisle of a four-aisle ASRS contains 40 storage compartments in the length direction and 10 compartments vertically. All storage compartments will be the same size to accommodate standard size pallets of dimensions: $x = 1$ m, $y = 1.2$ m. The height of a unit load $z = 0.9$ m. Using the allowances $a = 0.15$ m, $b = 0.2$ m, and $c = 0.25$ m, determine: (a) how many unit loads can be stored in the ASRS, and (b) the width, length, and height of the ASRS.

Solution:

The storage capacity: $\mathbf{CPA = 2 \times n_y \times n_z = 2 \times 40 \times 10 = 800}$ $\mathbf{TC = 4 \times 800 = 3200}$

The dimensions of the ASRS: $\mathbf{W = 3 \times (x + a) = 3 \times (1 + 0,15) = 3,45 \text{ m/aisle}}$

$$\mathbf{TW = 4 \times 3,45 = 13,8 \text{ m}}$$

$$\mathbf{L = n_y \times (y + b) = 40 \times (1,2 + 0,2) = 56 \text{ m}}$$

$$\mathbf{H = n_z \times (z + c) = 10 \times (0,9 + 0,25) = 11,5 \text{ m}}$$



Throughput of an ASRS (1)

System throughput is defined as the hourly rate of S/R transactions that the automated storage systems can perform.

- ❑ A transaction involve:
 - depositing a load into a storage,
 - retrieving a load from storage
- ❑ Transactions can be accomplished:
 - in a single command cycle,
 - in a dual command cycle – this one increase the throughput



For the throughput determination are considered:

- randomized storage of loads in the ASRS,
- storage compartments are equal size,
- the P&D station is located at the base and end of the aisle,
- constant horizontal and vertical speed of the S/R machine,
- simultaneous horizontal and vertical travel.

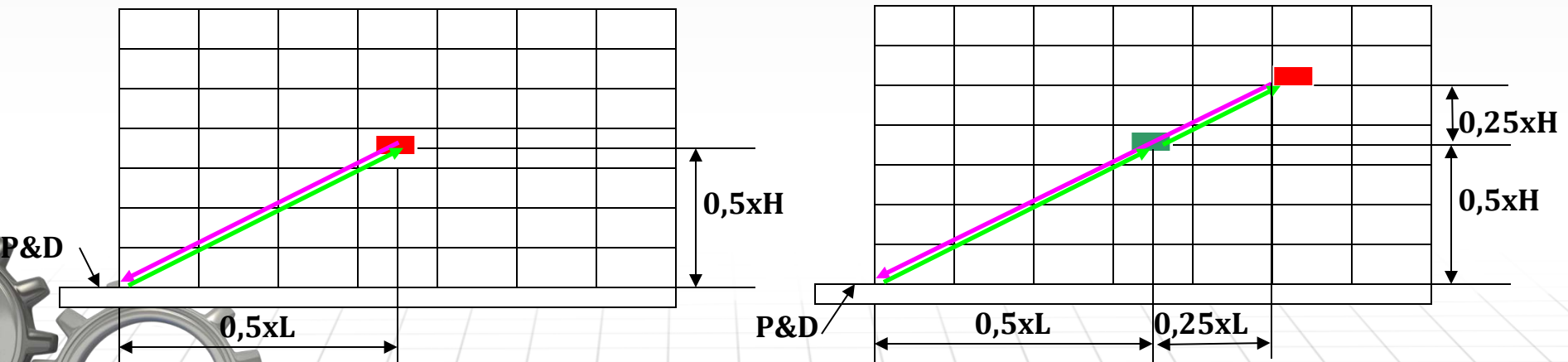
Throughput of an ASRS (2)

The single command cycle time can be expressed by:

$$T_{cs} = 2 \times \text{Max} \left\{ \frac{0,5 \times L}{v_y}, \frac{0,5 \times H}{v_z} \right\} + 2 \times T_{pd} = \text{Max} \left\{ \frac{L}{v_y}, \frac{H}{v_z} \right\} + 2 \times T_{pd}$$

The dual commands cycle time can be expressed by:

$$T_{cd} = 2 \times \text{Max} \left\{ \frac{0,75 \times L}{v_y}, \frac{0,75 \times H}{v_z} \right\} + 4 \times T_{pd} = \text{Max} \left\{ \frac{1,5 \times L}{v_y}, \frac{1,5 \times H}{v_z} \right\} + 4 \times T_{pd}$$



T_{cs} – cycle time of a single command cycle (min/cycle), T_{cd} – cycle time for a dual command cycle (min/cycle) L – length of the ASRS rack structure (m), v_y – velocity of the S/R machine along the length of the ASRS (m/min), H – height of the rack structure (m), v_z – velocity of the S/R machine in the vertical direction of the ASRS (m/min), T_{pd} – pickup-and-deposit time (min).



Throughput of an ASRS (3)

System throughput depends on the relative numbers of single and dual command cycles performed by the system.

The amounts of time spent in performing single command and dual command cycles each hour can be expressed by:

$$R_{cs} \times T_{cs} + R_{cd} \times T_{cd} = 60 \times U$$

The total hourly cycle rate is given by:

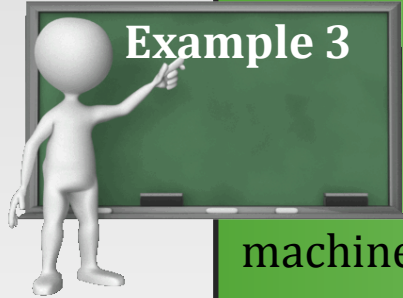
$$R_c = R_{cs} + R_{cd}$$

The total number of transactions performed per hour is:

$$R_t = R_{cs} + 2 \times R_{cd}$$

R_{cs} – number of single command cycles performed per hour; R_{cd} – number of dual command cycles performed per hour; U – system utilization during the hour; T_{cs} – cycle time of a single command cycle (min/cycle); T_{cd} – cycle time for a dual command cycle (min/cycle); R_c – total S/R cycle rate (cycles/hr); R_t – total number of transactions performed per hour.

Throughput of an ASRS



Example 3

Consider the ASRS from example 6, in which an S/R machine is used for each aisle. The length of the storage aisle is 56 m and its height is 11.5 m. Suppose horizontal and vertical speeds of the S/R machine are 60 m/min and 20 m/min, respectively. The S/R machine requires 20 sec to accomplish a P&D operation. Find: (a) the single command and dual command cycle times per aisle, and (b) throughput per aisle under the assumptions that storage system utilization is 90% and the number of single command and dual command cycles are equal.

Solution:

The single and dual command cycle times are given by:

$$T_{cs} = 2 \times \text{Max} \left\{ \frac{0,5 \times L}{v_y}, \frac{0,5 \times H}{v_z} \right\} + 2 \times T_{pd} = 2 \times \text{Max} \left\{ \frac{0,5 \times 56}{60}, \frac{0,5 \times 11,5}{20} \right\} + 2 \times \frac{20}{60} = 1,6 \text{ min/cycle}$$

$$T_{cd} = 2 \times \text{Max} \left\{ \frac{0,75 \times L}{v_y}, \frac{0,75 \times H}{v_z} \right\} + 4 \times T_{pd} = 2 \times \text{Max} \left\{ \frac{0,75 \times 56}{60}, \frac{0,75 \times 11,5}{20} \right\} + 4 \times \frac{20}{60} = 2,73 \text{ min./cycle}$$

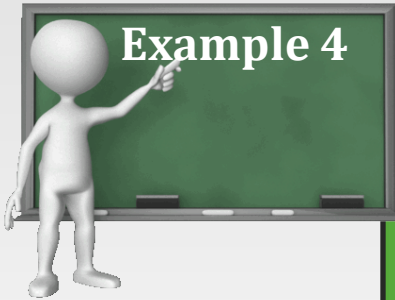
The throughput of the ASRS:

$$R_{cs} \times T_{cs} + R_{cd} \times T_{cd} = 60 \times U \Leftrightarrow R_{cs} \times 1,6 + R_{cd} \times 2,73 = 60 \times 0,9 = 54 \text{ min}$$

$$R_{cs} = R_{cd} \Rightarrow 4,33 \times R_{cs} = 54 \Rightarrow R_{cs} = R_{cd} = 12,47 \text{ cycles/hr}$$

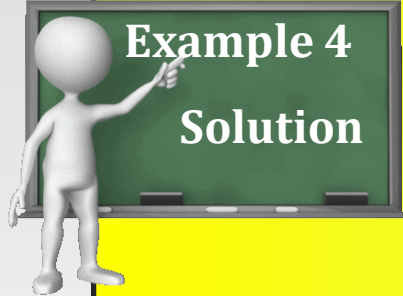
$$R_t = R_{cs} + 2 \times R_{cd} = 12,47 + 2 \times 12,47 = 37,41 \text{ transactions/hr}$$

Throughput of an ASRS



The aisles in the ASRS of the previous example will be organized following a class-based dedicated storage strategy. There will be two classes (A and B), according to activity level. The more-active stock is stored in the half of the rack system that is located closest to the input/output station, and the less-active stock is stored in the other half of the rack system farther away from the input/output station. Within each half of the rack system, random storage is used. The more-active stock accounts for 75% of the transactions, and the less-active stock accounts for the remaining 25%. As before, assume that system utilization is 90%, and the number of single command cycles and dual command cycles are equal. Determine the throughput of the ASRS.

Throughput of an ASRS



Example 4 Solution

$$T_{csA} = 2 \times \text{Max} \left\{ \frac{0.5 \times 28}{60}, \frac{0.5 \times 11.5}{20} \right\} + 2 \times \left(\frac{20}{60} \right) = 1.241 \text{ min}$$

$$T_{cdA} = 2 \times \text{Max} \left\{ \frac{0.75 \times 28}{60}, \frac{0.75 \times 11.5}{20} \right\} + 4 \times \left(\frac{20}{60} \right) = 2.196 \text{ min}$$

$$T_{csB} = 2 \times \text{Max} \left\{ \frac{28 + 0.5 \times 28}{60}, \frac{0.5 \times 11.5}{20} \right\} + 2 \times \left(\frac{20}{60} \right) = 2.067 \text{ min}$$

$$T_{cdB} = 2 \times \text{Max} \left\{ \frac{28 + 0.75 \times 28}{60}, \frac{0.75 \times 11.5}{20} \right\} + 4 \times \left(\frac{20}{60} \right) = 2.966 \text{ min}$$

$$R_{csA} = R_{cdA} \text{ and } R_{csB} = R_{cdB} \quad R_{csA} = 3 \times R_{csB} \text{ and } R_{cdA} = 3 \times R_{cdB}$$

$$R_{csA} \times T_{csA} + R_{cdA} \times T_{cdA} + R_{csB} \times T_{csB} + R_{cdB} \times T_{cdB} = 60 \times 0.9$$

$$R_{csA} \times T_{csA} + R_{csA} \times T_{cdA} + R_{csB} \times T_{csB} + R_{csB} \times T_{cdB} = 60 \times 0.9$$

$$3 \times R_{csB} \times T_{csA} + 3 \times R_{csB} \times T_{cdA} + R_{csB} \times T_{csB} + R_{csB} \times T_{cdB} = 60 \times 0.9$$

$$3 \times R_{csB} \times 1.241 + 3 \times R_{csB} \times 2.196 + R_{csB} \times 2.067 + R_{csB} \times 2.966 = 54$$

$$15.344 \times R_{csB} = 54 \Rightarrow R_{csB} = 3.519 \quad R_{csA} = 3 \times R_{csB} = 10.557$$

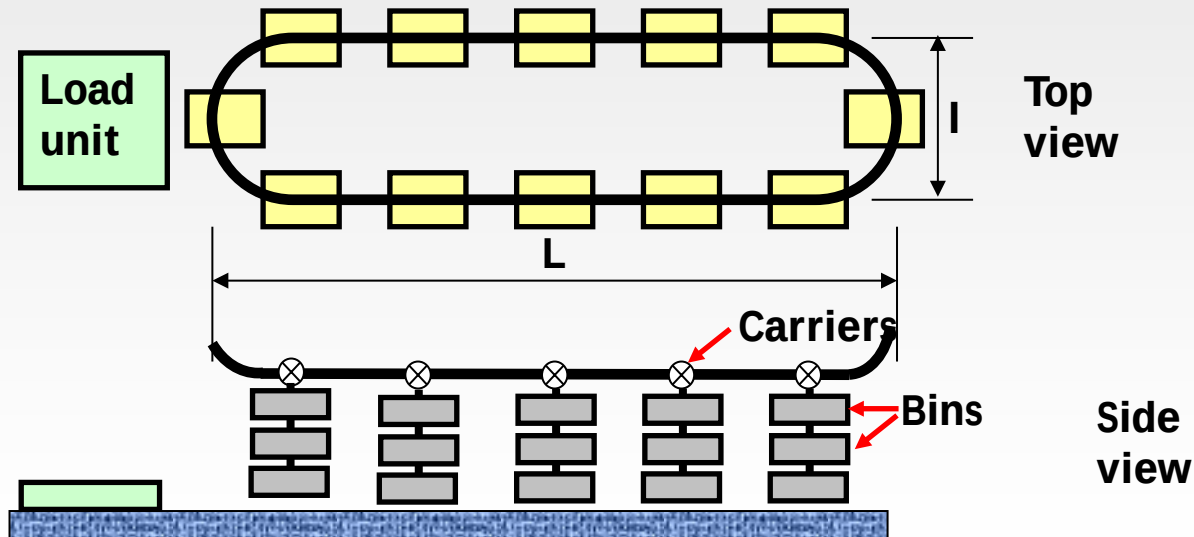
$$R_{cdB} = R_{csB} = 3.519 \quad R_{cdA} = 3 \times R_{cdB} = 10.557$$

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$$R_t = R_{csA} + R_{csB} + 2 \times (R_{cdA} + R_{cdB}) = 10,557 + 3,519 + 2 \times (10,557 + 3,519) = 42,23 \text{ transactions/hr}$$

Storage capacity of a carousel

Top and side views of a horizontal carousel with 12 carriers ($n_c=12$) and 3 bins/carrier ($n_b=3$)



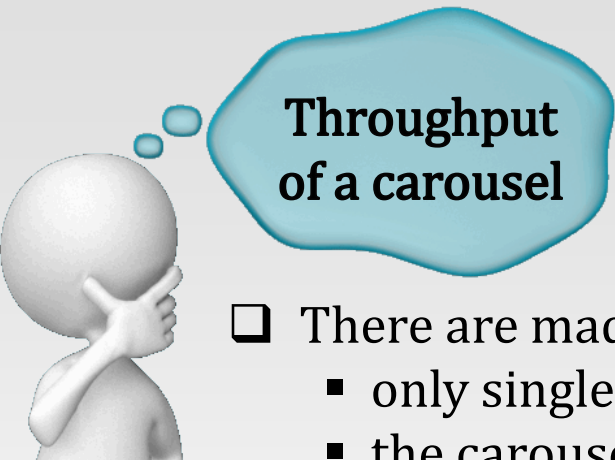
The circumference of the rail is given by: $C = 2 \times (L - W) + \pi \times W$

Total number of bins: $N = n_c \times n_b$

The carriers are separated by a distance:

$$C = s_c \times n_c$$

where s_c - the center to center spacing of carriers



Throughput of a carousel

- ❑ There are made some assumptions:
 - only single command cycles are performed,
 - the carousel operate with a constant speed v_c ,
 - random storage is assumed,
 - the carousel can move in either direction.
- ❑ The mean travel distance between the load/unload station and a bin randomly located in the carousel is $C/4$.
- ❑ The S/R cycle time is given by:
- ❑ The number of transactions accomplished per hour is the same as the number of cycles and is given by:

$$T_c = \frac{C}{4 \times v_c} + T_{pd}$$

$$R_t = R_c = \frac{60}{T_c}$$

T_c – S/R cycle time (min); C – carousel circumference (m); v_c – carousel velocity (m/min); T_{pd} – average time required to pick/deposit items each cycle by the operator at the load/unload station (min); R_t – number of transactions; R_c – number of cycles

Carousel operation



Example 5

The oval rail of a carousel storage system has length=4 m, and width=0.5 m. There are 12 carriers equally spaced around the oval. Suspended from each carrier are 3 bins. Each bin has a volumetric capacity=0.01 m³. Carousel speed =20 m/min. Average P&D time for a retrieval=20 sec.. Determine: (a) volumetric capacity of the storage system and (b) hourly retrieval rate of the storage system.

Solution:

Total number of bins in the carousel is: $N = n_c \times n_b = 12 \times 3 = 36$

Total volumetric capacity = $36 \times 0,01 = 0,36 \text{ m}^3$

The circumference of the carousel rail: $C = 2 \times (4 - 0,5) + \pi \times 0,5 = 8,57 \text{ m}$

Cycle time per retrieval: $T_c = \frac{8,57}{4 \times 20} + \frac{20}{60} = 0,437 \text{ min}$

The throughput expressed as an hourly rate is:

$$R_t = \frac{60}{0,437} = 137,26 \text{ retrieval transactions/hr}$$



Self-assessment test

1. Usually in a storage system the accessibility increases as the density increases.

T

F

2. Availability of ASRS is a measure of its reliability.

T

F

3. Random storage is used when products are not stored according to the level of activity.

T

F

4. In the case of a carousel storage system, its robot arm moves to each basket.

T

F

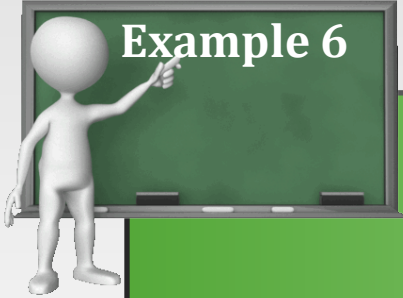
5. An ASRS is more efficient when performing in single operation mode than in dual operation mode.

T

F

Material Handling Systems

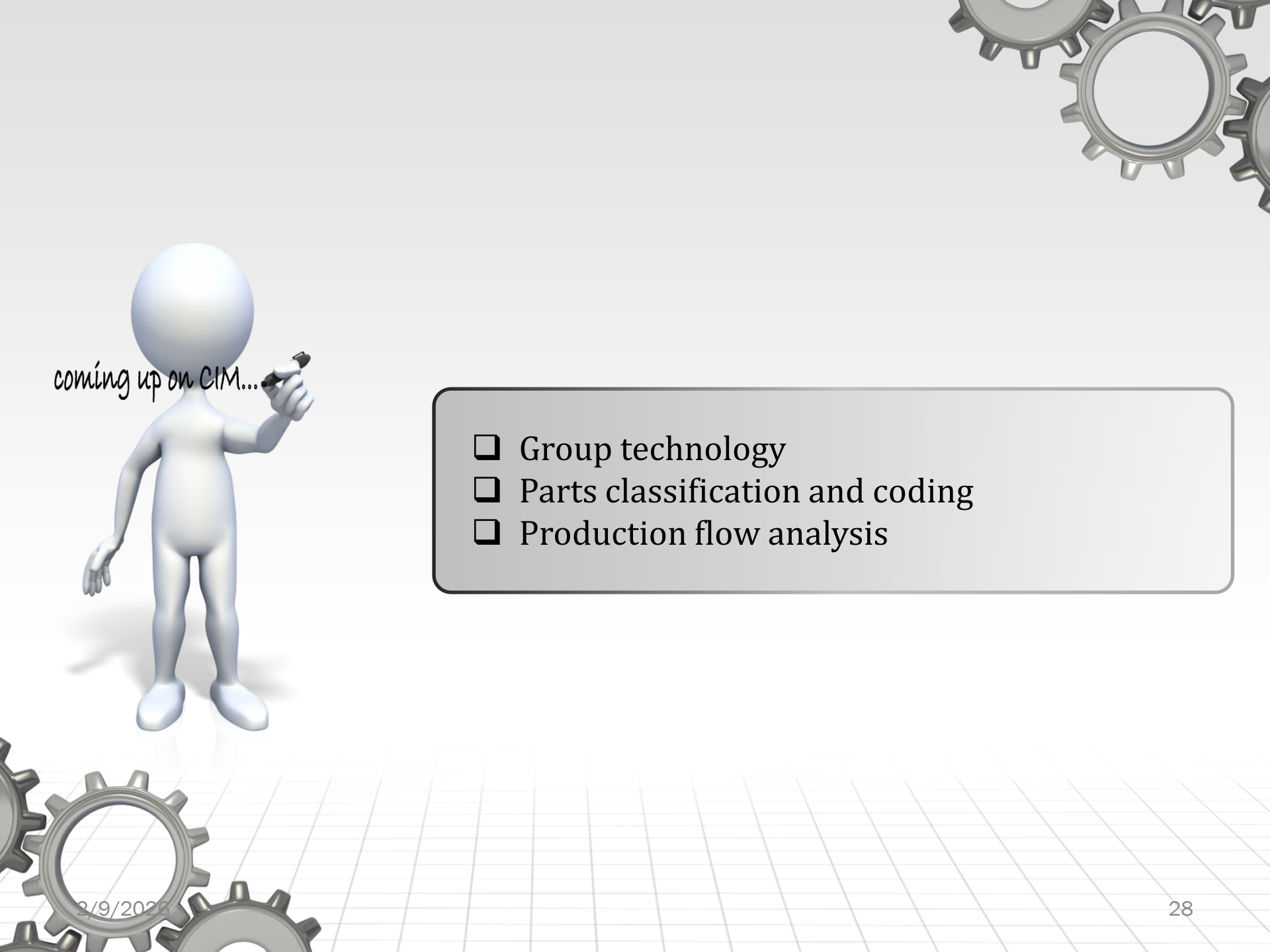
Example 6





Conclusion

- ☐ Material handling must be done in compliance with the 10 principles.
- ☐ The choice of the storage modality, depends on the nature of materials and on the manufacturing system configuration.
- ☐ The storage strategy should not negatively affect the efficiency of the production system.
- ☐ To determine the efficiency of a storage system, it always starts from some assumptions.



coming up on CIM...

- ☐ Group technology
- ☐ Parts classification and coding
- ☐ Production flow analysis